

PRINCIPLES OF ECOLOGICAL ECONOMICS

HEKA22

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Learning Objectives

 By the end of today's lecture you should:

1. Be able to describe how ecological economics relates economic, ecological, and social systems
2. Be able to list ways in which ecological economics challenges mainstream economic thinking
3. Be able to criticize mainstream approaches to value nature from the perspective of strong sustainability

Principles

**Ecological Economics embeds the economy in society
and society in the environment**

Empty World

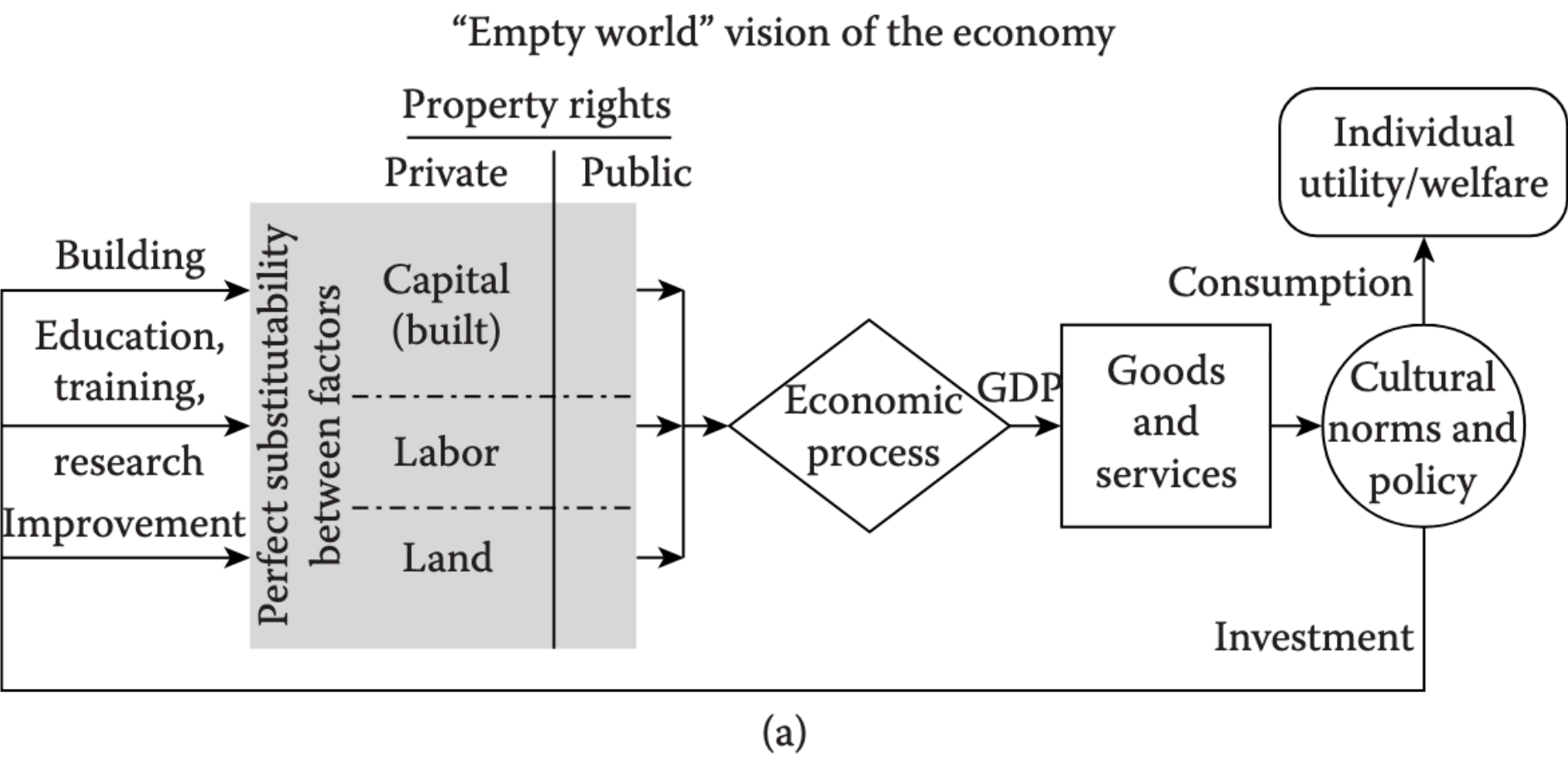


Figure 1: Stylistic Image of Empty World Economics copied from Costanza (2015, 159)

How many economists think about provisioning: Cobb-Douglas Production Function

A combination of three “production factors”

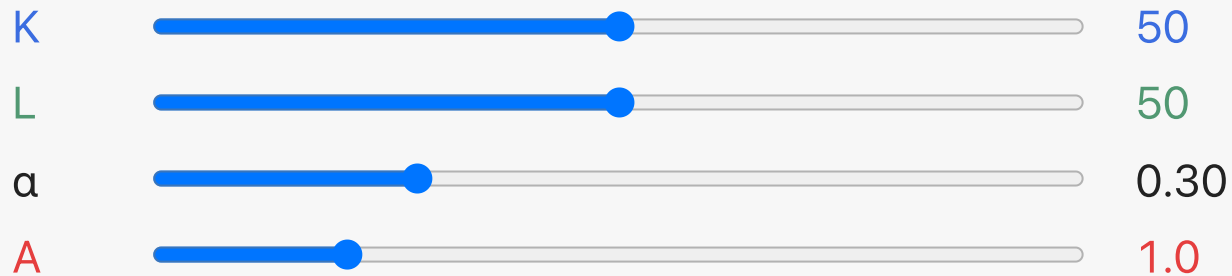
$$Y = A \cdot K^{\alpha} \cdot L^{1-\alpha}$$

With:

A = Productivity, K = capital, L = labor, $\alpha + (1 - \alpha) = 1$

Building some intuition for the Cobb-Douglas production function

$$Y = A \cdot K^{\alpha} \cdot L^{1-\alpha}$$



$$Y = 50.00$$

Full-World Economics

Criticism from Georgescu-Roegen (1975)

Strong versus Weak Sustainability

Ecosystem services revisited

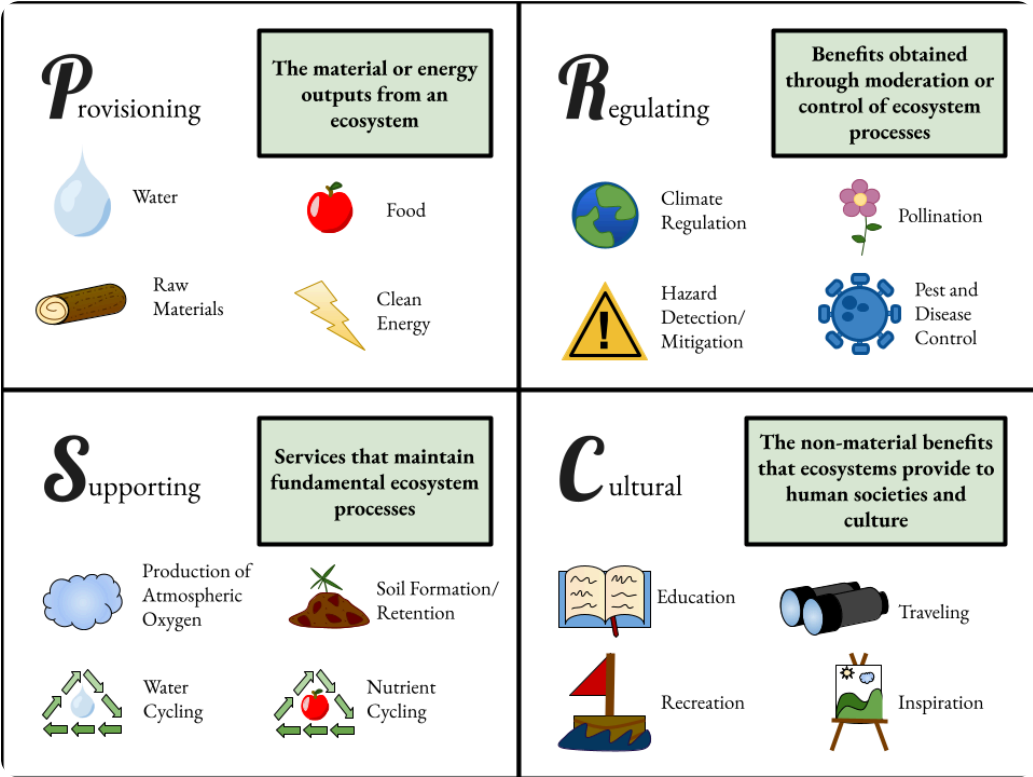


Figure 2: Types of Ecosystem Services By Snoopdeeboop CC0, <https://commons.wikimedia.org/w/index.php?curid=148837942>

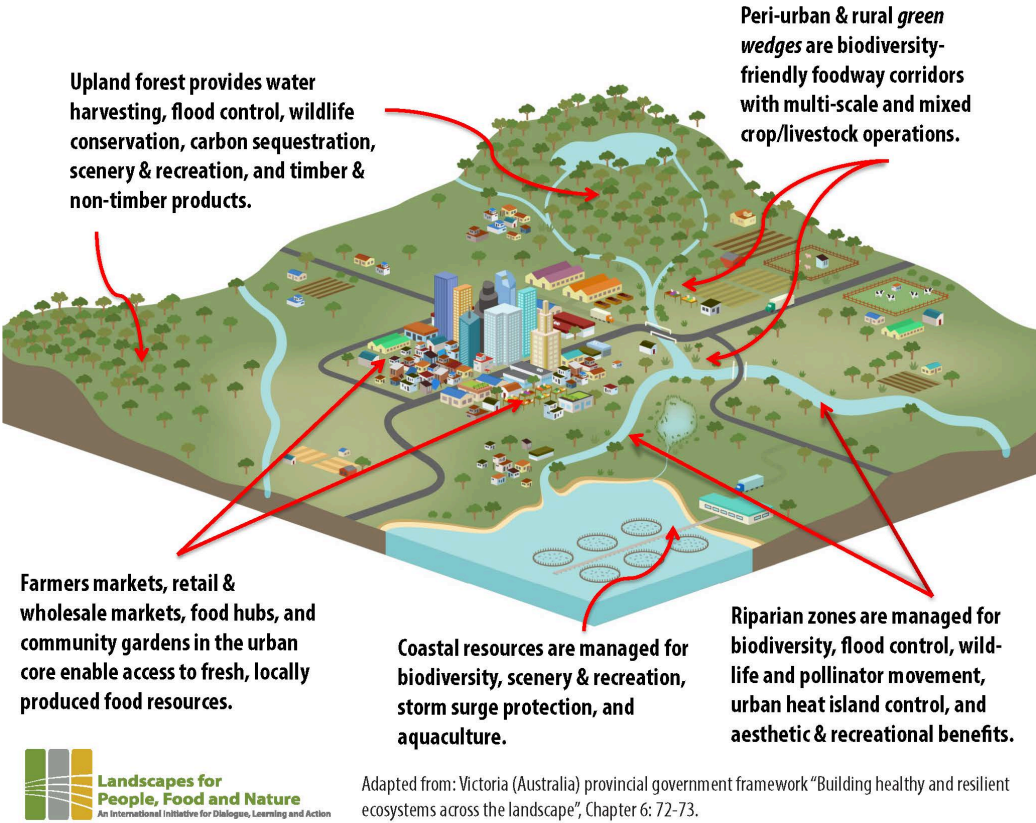


Figure 3: Examples of Ecosystem Services FORSTER, Thomas; ESCUDERO

Scale, distribution, and allocation

Seven ways to think like an Ecological Economist

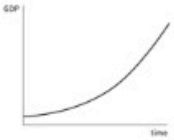
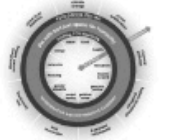
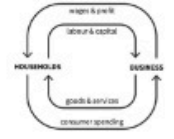
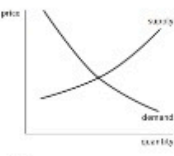
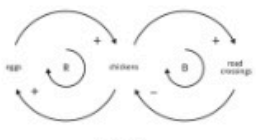
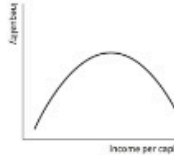
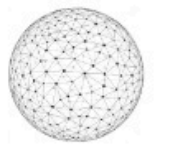
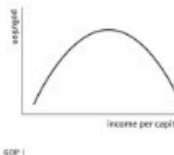
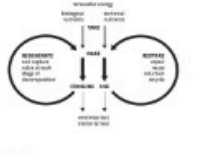
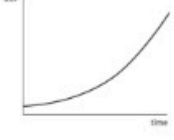
7 ways to think:	from 20 th century economics	to 21 st century economics
1. Change the goal	 GDP	 the Doughnut
2. See the big picture	 self-contained market	 embedded economy
3. Nurture human nature	 rational economic man	 social adaptable humans
4. Get savvy with systems	 mechanical equilibrium	 dynamic complexity
5. Design to distribute	 growth will even it up again	 distributive by design
6. Create to regenerate	 growth will clean it up again	 regenerative by design
7. Be agnostic about growth	 growth addicted	 growth agnostic

Figure 4: Kate Raworth (2018)'s Seven Ways to Think Like an Economist for the 21st Century

Checking-in on Learning Objectives

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References

Costanza, Robert. 2015. *An Introduction to Ecological Economics*. Second edition. CRC Press, Taylor & Francis Group.

Georgescu-Roegen, Nicholas. 1975. "Energy and Economic Myths." *Southern Economic Journal* 41 (3): 347.

<https://doi.org/10.2307/1056148>.

Raworth, Kate. 2018. *Doughnut Economics: Seven Ways to Think Like a 21st-Century Economist*. Paperback edition. Random House Business Books.